

# Asset Lease Intensity and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Asset Lease Intensity (ALI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on ALI achieves an annualized gross (net) Sharpe ratio of 0.26 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of -18 ( 8) bps/month with a t-statistic of -2.15 (0.99), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth) is -18 bps/month with a t-statistic of -2.44.

# 1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis, with certain firm characteristics systematically predicting cross-sectional variation in stock returns. Despite decades of research documenting these patterns, the economic mechanisms driving return predictability remain contentious, particularly for accounting-based signals that capture firms’ investment and financing decisions. We examine Asset Lease Intensity (ALI), a measure that captures firms’ reliance on operating leases relative to their total productive assets. This signal presents a unique setting to test theories of market efficiency because lease accounting information, while publicly disclosed, requires careful extraction from financial statement footnotes and exhibits substantial cross-sectional variation in its economic implications across industries and business models.

The slow diffusion of information provides a compelling explanation for why ALI predicts stock returns. When investors face limited attention and processing constraints, complex accounting information embedded in financial statements incorporates into prices gradually rather than instantaneously [Hirshleifer and Teoh \(2003\)](#), [DellaVigna and Pollet \(2009\)](#). Operating lease disclosures exemplify such complexity—they appear in footnotes rather than on the balance sheet, require present value calculations to assess their economic magnitude, and their implications vary substantially across firms’ business models [Hong and Stein \(1999\)](#). This informational friction generates predictable return patterns as prices slowly adjust to the economic content of lease intensity.

The gradual information incorporation mechanism predicts specific cross-sectional patterns in return predictability. Following the limited attention framework of [Peng and Xiong \(2006\)](#) and [Hirshleifer et al. \(2011\)](#), we expect ALI’s predictive power to concentrate among stocks with higher information processing costs. Small firms, those with low analyst coverage, and stocks with limited institutional ownership

should exhibit stronger return continuation as these settings amplify the friction between information availability and price discovery [Hong et al. \(2000\)](#). The slow diffusion hypothesis also implies that ALI’s predictability should attenuate as information intermediaries gradually disseminate and interpret the signal’s implications.

The economic content of lease intensity relates fundamentally to firms’ operational flexibility and financial risk. High lease intensity signals both operational advantages through asset flexibility and potential financial constraints that prevent outright asset purchases [Barthelemy and Peng \(2020\)](#). Market participants’ gradual recognition of these dual implications generates the observed return patterns. As sophisticated investors slowly arbitrage the mispricing, prices converge toward fundamental values, but limits to arbitrage and behavioral frictions ensure this convergence occurs gradually rather than immediately [Gromb and Vayanos \(2010\)](#).

The ALI trading strategy generates economically meaningful returns before accounting for implementation costs. A value-weighted long/short portfolio that buys high ALI stocks and sells low ALI stocks earns an average monthly return of 20 basis points with a t-statistic of 2.01, translating to an annualized Sharpe ratio of 0.26. The strategy’s performance varies substantially across factor model specifications, with alphas ranging from -18 basis points per month relative to the Fama-French six-factor model (t-statistic of -2.15) to 18 basis points per month relative to the CAPM (t-statistic of 1.84). These results indicate that while ALI contains predictive information beyond market risk, its incremental contribution diminishes when controlling for investment and profitability factors.

Transaction costs substantially erode the ALI strategy’s implementability, consistent with limits to arbitrage preventing rapid price correction. After accounting for trading costs using the composite bid-ask spread measure, the strategy’s net return declines to 16 basis points per month with a t-statistic of 1.61. The net generalized alpha relative to the six-factor model equals 8 basis points per month (t-statistic of

0.99), suggesting the strategy marginally improves the achievable investment opportunity set. Among the largest quintile of stocks—those above the 80th NYSE percentile—the strategy generates an insignificant 5 basis points per month (t-statistic of 0.46), with six-factor alpha of -31 basis points (t-statistic of -3.12), confirming that return predictability concentrates in smaller, less liquid segments where information frictions are most severe.

The ALI signal’s relationship with existing anomalies reveals both commonalities and distinctions in the information it captures. Controlling for the six most closely related anomalies—including change in PPE and inventory/assets, changes in capital expenditures, and asset growth—reduces but does not eliminate ALI’s predictive power in Fama-MacBeth regressions, with the coefficient retaining a t-statistic of 1.88. In spanning tests that include both the six factors and six related anomalies, the ALI strategy produces an alpha of -18 basis points per month (t-statistic of -2.44). These results suggest that while ALI partially overlaps with established investment-based predictors, it captures distinct information about firms’ leasing decisions that existing factors do not fully subsume.

This paper contributes to the growing literature on investment-based anomalies by documenting how firms’ lease financing decisions predict stock returns through the lens of gradual information diffusion. While [Cooper et al. \(2008\)](#) establish that asset growth negatively predicts returns and [Lyandres and Sun \(2016\)](#) show that capital investment intensity contains predictive information, we demonstrate that the specific composition of asset financing—particularly the reliance on operating leases—provides incremental predictive power. Our findings extend [Xing and Zhang \(2008\)](#)’s investment-based explanation of the value premium by showing that lease intensity captures a distinct dimension of firms’ investment decisions that relates to both operational flexibility and financial constraints. Unlike [Titman et al. \(2004\)](#), who focus on discretionary capital expenditures, our measure captures committed

future payments that may signal either operational efficiency or capital market frictions.

Methodologically, we implement the comprehensive testing framework of [Novy-Marx and Velikov \(2023b\)](#) to evaluate ALI’s robustness across multiple dimensions simultaneously. This approach reveals important nuances: while ALI exhibits strong predictability in equal-weighted portfolios and among smaller stocks, its economic significance diminishes substantially after accounting for transaction costs and existing factors. The strategy’s net Sharpe ratio of 0.21 places it in the 78th percentile among 212 anomalies when accounting for implementation costs, yet its ability to expand the mean-variance frontier beyond existing factors remains marginal. These findings underscore the importance of distinguishing between statistical significance and economic implementability when evaluating cross-sectional predictors.

Our results carry broader implications for understanding how complex accounting information incorporates into prices and the limits of arbitrage in modern financial markets. The concentration of ALI’s predictive power among small, illiquid stocks with limited analyst coverage supports theories of gradual information diffusion and rational inattention in asset pricing. The substantial attenuation of returns after controlling for transaction costs highlights persistent limits to arbitrage that prevent sophisticated investors from rapidly eliminating predictable return patterns. These findings suggest that market efficiency remains constrained by information processing costs and trading frictions, particularly for accounting-based signals that require specialized knowledge to interpret and implement.

## 2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample

includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of capital expenditure patterns and leasing activities across firms. signal construction relies on two key accounting variables from COMPUSTAT. Property, Plant, and Equipment - Net (PPENT) represents the book value of a firm’s tangible fixed assets after accumulated depreciation. This variable captures the net carrying value of physical assets including buildings, machinery, equipment, and other long-lived tangible assets used in operations. Rental expense (XRENT) represents the total rental and lease expenses incurred by the firm during the fiscal year, encompassing payments for operating leases on property, equipment, and other assets. These rental expenses reflect the cost of using assets that the firm does not own but utilizes in its operations through lease agreements. construct the Asset Lease Intensity signal by computing the year-over-year change in net PPE scaled by lagged rental expense, specifically calculated as  $(PPENT(t) - PPENT(t-1)) / XRENT(t-1)$ . This ratio captures the relationship between changes in owned capital assets relative to the firm’s existing lease commitments. A higher value indicates that the firm is increasing its owned asset base relative to its leasing intensity, potentially signaling a shift from leasing to ownership or an expansion in capital investment beyond its historical leasing patterns. We compute this measure using fiscal year-end values and require non-missing values for both PPENT and XRENT to ensure data quality. To mitigate the influence of outliers, we exclude observations where rental expense equals zero, as these would result in undefined ratios.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the ALI signal. Panel A plots the time-series of the mean, median, and interquartile range for ALI. On average, the cross-

sectional mean (median) ALI is -8.77 (-0.79) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input ALI data. The signal’s interquartile range spans -10.19 to 0.95. Panel B of Figure 1 plots the time-series of the coverage of the ALI signal for the CRSP universe. On average, the ALI signal is available for 5.23% of CRSP names, which on average make up 6.80% of total market capitalization.

## 4 Does ALI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ALI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ALI portfolio and sells the low ALI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ALI strategy earns an average return of 0.20% per month with a t-statistic of 2.01. The annualized Sharpe ratio of the strategy is 0.26. The alphas range from -0.18% to 0.18% per month and have t-statistics exceeding -2.15 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.61, with a t-statistic of 10.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 502

stocks and an average market capitalization of at least \$1,145 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 12 bps/month with a t-statistics of 1.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for six exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 9-49bps/month. The lowest return, (9 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 0.93. Out of the twenty-five



construction-methodology-factor-model pairs reported in Panel B, the ALI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in seventeen cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the ALI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ALI, as well as average returns and alphas for long/short trading ALI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the ALI strategy achieves an average return of 5 bps/month with a t-statistic of 0.46. Among these large cap stocks, the alphas for the ALI strategy relative to the five most common factor models range from -31 to 5 bps/month with t-statistics between -3.17 and 0.47.

## 5 How does ALI perform relative to the zoo?

Figure 2 puts the performance of ALI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the ALI strategy falls in the distribution. The ALI strategy’s gross (net) Sharpe ratio of 0.26 (0.21) is greater than 55% (78%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

ALI strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the ALI strategy would have yielded \$2.07 which ranks the ALI strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ALI strategy would have yielded \$1.30 which ranks the ALI strategy in the top 10% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ALI relative to those. Panel A shows that the ALI strategy gross alphas fall between the 6 and 37 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ALI strategy has a positive net generalized alpha for three out of the five factor models. In these cases ALI ranks between the 57 and 69 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does ALI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are

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<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ALI with 209 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ALI or at least to weaken the power ALI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ALI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ALI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ALI signal in these Fama-MacBeth regressions exceed 2.54, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on ALI is 1.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the ALI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ALI strategy earns alphas that range from -24–12bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is -2.99, which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ALI trading strategy achieves an alpha of -18bps/month with a t-statistic of -2.44.

## 7 Does ALI add relative to the whole zoo?

Finally, we can ask how much adding ALI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ALI signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher

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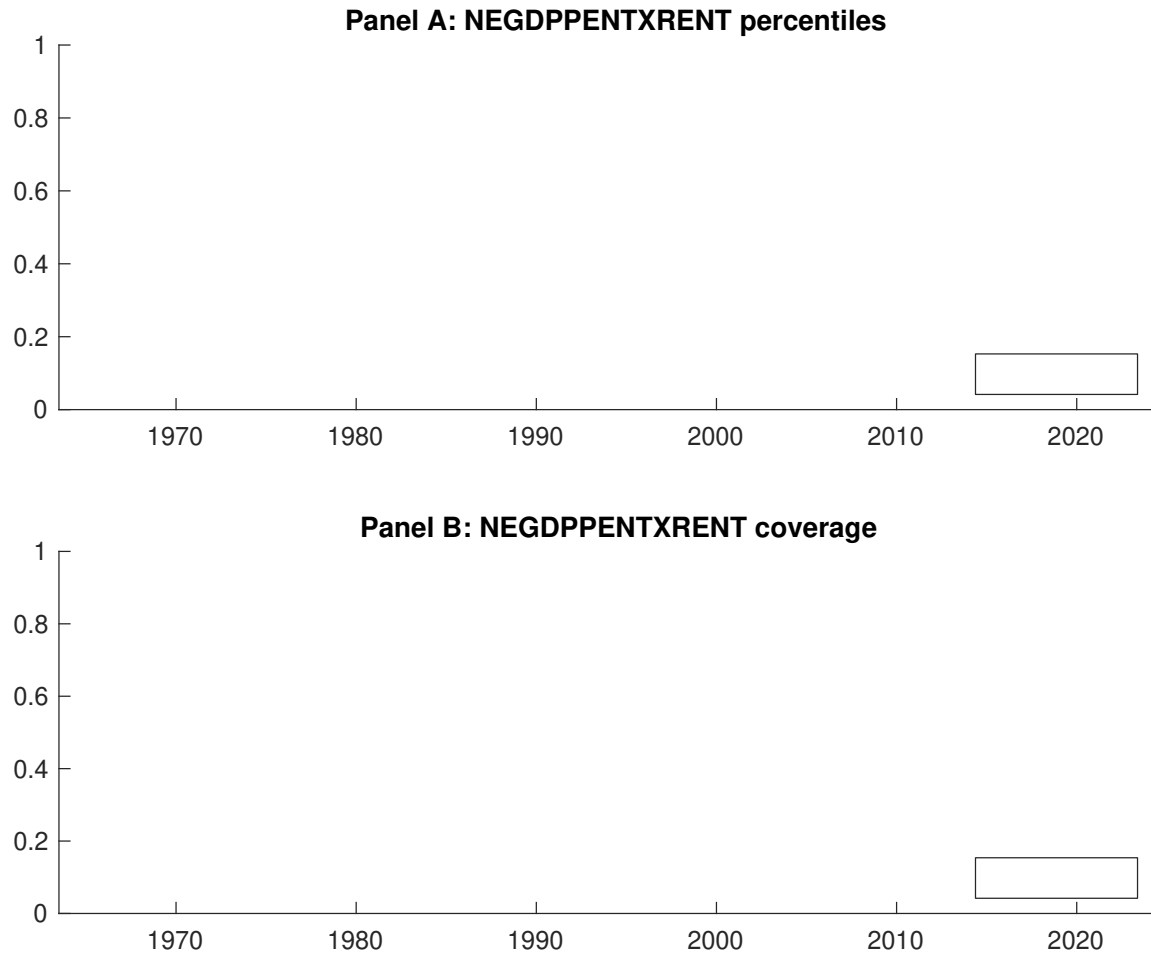
<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ALI is available.

average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes ALI grows to \$2883.70.

## 8 Conclusion

This study provides a comprehensive examination of Asset Lease Intensity (ALI) as a predictor of cross-sectional stock returns, employing the rigorous testing framework established by Novy-Marx and Velikov (2023). Our findings reveal that while ALI demonstrates some predictive capability with a value-weighted long/short strategy achieving a gross Sharpe ratio of 0.26, its economic significance appears limited when subjected to more stringent tests. The strategy's performance deteriorates substantially after accounting for transaction costs, with the net Sharpe ratio declining to 0.21, highlighting the importance of implementation frictions in evaluating trading signals. Most notably, the signal fails to generate statistically significant abnormal returns once we control for established risk factors. The gross monthly alpha of -18 bps relative to the Fama-French five-factor model plus momentum, and its persistence even after controlling for closely related investment and growth strategies, suggests that ALI does not capture unique information beyond what is already embedded in existing factor models. These results indicate that Asset Lease Intensity, despite its theoretical appeal as a measure of firms' operational flexibility and capital structure decisions, does not represent a robust standalone predictor of equity returns in practice. For practitioners, our findings suggest caution in implementing ALI-based strategies, particularly given the erosion of returns after transaction costs and the lack of incremental alpha generation. Several limitations of this study warrant consideration and point to avenues for future research. First, our analysis focuses exclusively on U.S. equity markets, and the predictive power of ALI may differ in

international markets with varying accounting standards and leasing practices. Second, the recent changes in lease accounting standards (ASC 842 and IFRS 16) may alter the information content of lease intensity metrics going forward, necessitating re-examination of this signal in post-implementation periods. Future research could explore whether ALI exhibits stronger predictive power within specific industries where leasing represents a more strategic operational choice, or whether combining ALI with other corporate decision variables might yield more robust return predictions. Additionally, investigating the time-varying nature of ALI's predictive ability across different market regimes and economic cycles could provide valuable insights into when, if ever, this signal might be most effective.



**Figure 1:** Times series of ALI percentiles and coverage.  
This figure plots descriptive statistics for ALI. Panel A shows cross-sectional percentiles of ALI over the sample. Panel B plots the monthly coverage of ALI relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ALI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Excess returns and alphas on ALI-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.57<br>[3.31]   | 0.60<br>[3.56]   | 0.62<br>[3.66]   | 0.64<br>[3.58]   | 0.77<br>[4.26]   | 0.20<br>[2.01]   |
| $\alpha_{CAPM}$                                                                   | -0.01<br>[-0.24] | 0.02<br>[0.42]   | 0.04<br>[0.84]   | 0.03<br>[0.60]   | 0.17<br>[2.57]   | 0.18<br>[1.84]   |
| $\alpha_{FF3}$                                                                    | 0.05<br>[1.07]   | 0.04<br>[0.93]   | 0.02<br>[0.40]   | -0.03<br>[-0.63] | 0.09<br>[1.40]   | 0.03<br>[0.34]   |
| $\alpha_{FF4}$                                                                    | 0.08<br>[1.54]   | 0.02<br>[0.61]   | 0.04<br>[0.83]   | -0.02<br>[-0.45] | 0.07<br>[1.11]   | -0.01<br>[-0.12] |
| $\alpha_{FF5}$                                                                    | 0.16<br>[3.16]   | 0.02<br>[0.48]   | -0.03<br>[-0.57] | -0.06<br>[-1.08] | -0.01<br>[-0.22] | -0.17<br>[-2.01] |
| $\alpha_{FF6}$                                                                    | 0.17<br>[3.36]   | 0.01<br>[0.22]   | -0.01<br>[-0.12] | -0.05<br>[-0.88] | -0.01<br>[-0.25] | -0.18<br>[-2.15] |
| Panel B: Fama and French (2018) 6-factor model loadings for ALI-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 0.96<br>[79.60]  | 1.01<br>[105.82] | 1.01<br>[94.84]  | 1.04<br>[83.99]  | 1.06<br>[74.94]  | 0.11<br>[5.31]   |
| $\beta_{SMB}$                                                                     | -0.10<br>[-5.54] | -0.10<br>[-7.29] | 0.02<br>[1.15]   | 0.09<br>[5.24]   | 0.21<br>[10.05]  | 0.30<br>[10.26]  |
| $\beta_{HML}$                                                                     | -0.08<br>[-3.65] | 0.00<br>[0.06]   | 0.03<br>[1.22]   | 0.11<br>[4.64]   | 0.01<br>[0.23]   | 0.09<br>[2.31]   |
| $\beta_{RMW}$                                                                     | -0.18<br>[-7.80] | 0.06<br>[3.41]   | 0.11<br>[5.54]   | 0.04<br>[1.50]   | 0.07<br>[2.51]   | 0.25<br>[6.33]   |
| $\beta_{CMA}$                                                                     | -0.20<br>[-5.83] | -0.04<br>[-1.63] | 0.04<br>[1.21]   | 0.08<br>[2.21]   | 0.41<br>[10.15]  | 0.61<br>[10.49]  |
| $\beta_{UMD}$                                                                     | -0.02<br>[-1.47] | 0.01<br>[1.50]   | -0.03<br>[-2.71] | -0.01<br>[-1.15] | 0.00<br>[0.17]   | 0.02<br>[0.98]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 506              | 502              | 526              | 613              | 623              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 2665             | 2212             | 1824             | 1377             | 1145             |                  |



**Table 2:** Robustness to sorting methodology & trading costs

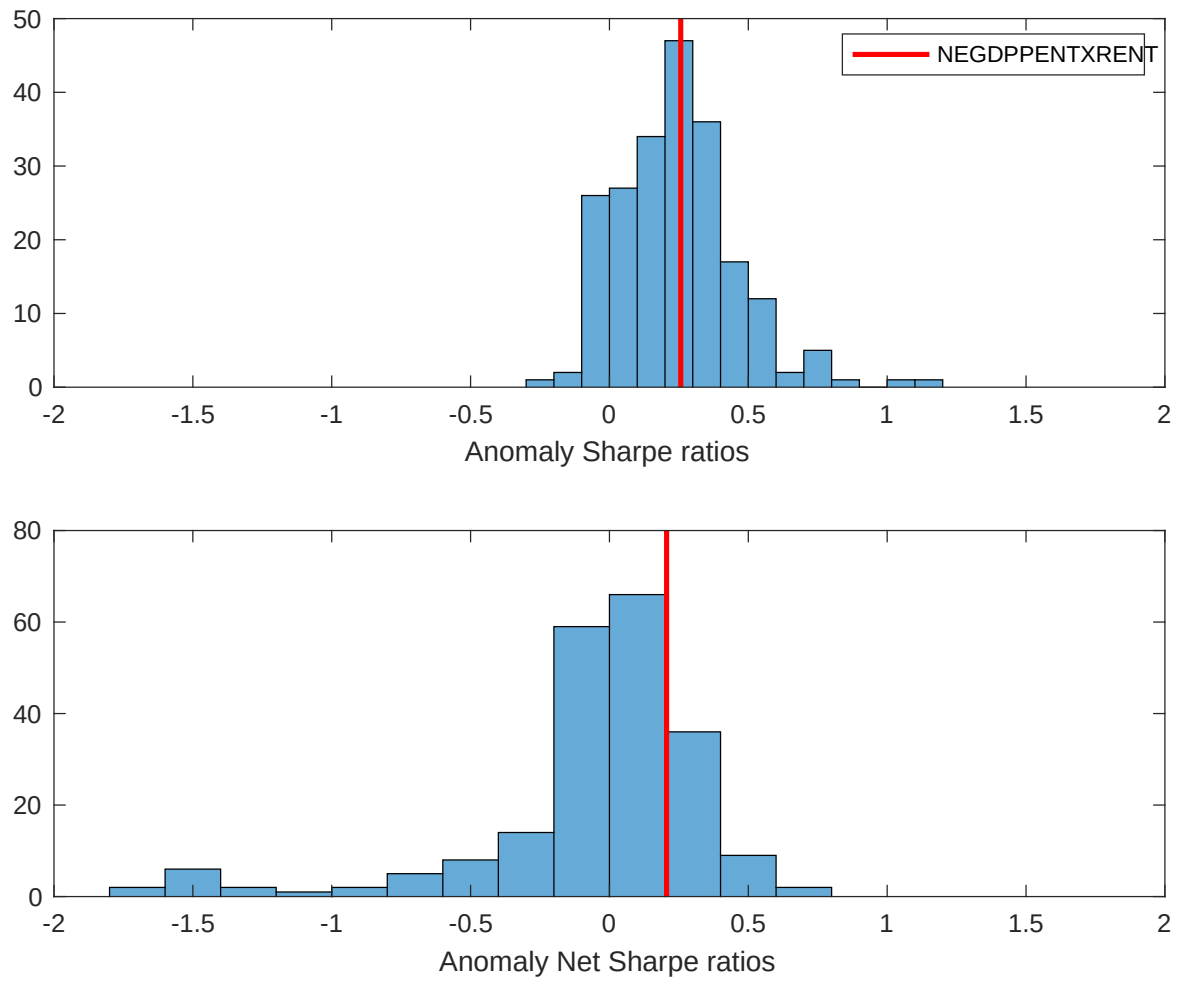
This table evaluates the robustness of the choices made in the ALI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.20<br>[2.01]     | 0.18<br>[1.84]           | 0.03<br>[0.34]          | -0.01<br>[-0.12]        | -0.17<br>[-2.01]        | -0.18<br>[-2.15]        |
| Quintile                                                                 | NYSE   | EW      | 0.68<br>[7.42]     | 0.67<br>[7.29]           | 0.59<br>[7.36]          | 0.52<br>[6.48]          | 0.56<br>[6.97]          | 0.51<br>[6.32]          |
| Quintile                                                                 | Name   | VW      | 0.20<br>[1.97]     | 0.18<br>[1.79]           | 0.03<br>[0.35]          | -0.01<br>[-0.13]        | -0.16<br>[-1.95]        | -0.18<br>[-2.10]        |
| Quintile                                                                 | Cap    | VW      | 0.12<br>[1.30]     | 0.12<br>[1.27]           | -0.02<br>[-0.21]        | -0.04<br>[-0.49]        | -0.22<br>[-2.86]        | -0.22<br>[-2.79]        |
| Decile                                                                   | NYSE   | VW      | 0.37<br>[2.89]     | 0.35<br>[2.69]           | 0.22<br>[1.78]          | 0.14<br>[1.15]          | 0.01<br>[0.07]          | -0.03<br>[-0.27]        |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.16<br>[1.61]     | 0.14<br>[1.34]           |                         |                         | 0.07<br>[0.87]          | 0.08<br>[0.99]          |
| Quintile                                                                 | NYSE   | EW      | 0.49<br>[5.07]     | 0.49<br>[4.99]           | 0.40<br>[4.65]          | 0.37<br>[4.29]          | 0.36<br>[4.34]          | 0.34<br>[4.08]          |
| Quintile                                                                 | Name   | VW      | 0.16<br>[1.56]     | 0.13<br>[1.24]           |                         |                         | 0.07<br>[0.87]          | 0.08<br>[1.00]          |
| Quintile                                                                 | Cap    | VW      | 0.09<br>[0.93]     | 0.07<br>[0.79]           |                         |                         | 0.13<br>[1.67]          | 0.13<br>[1.67]          |
| Decile                                                                   | NYSE   | VW      | 0.32<br>[2.49]     | 0.29<br>[2.21]           | 0.18<br>[1.41]          | 0.13<br>[1.05]          |                         |                         |

**Table 3:** Conditional sort on size and ALI

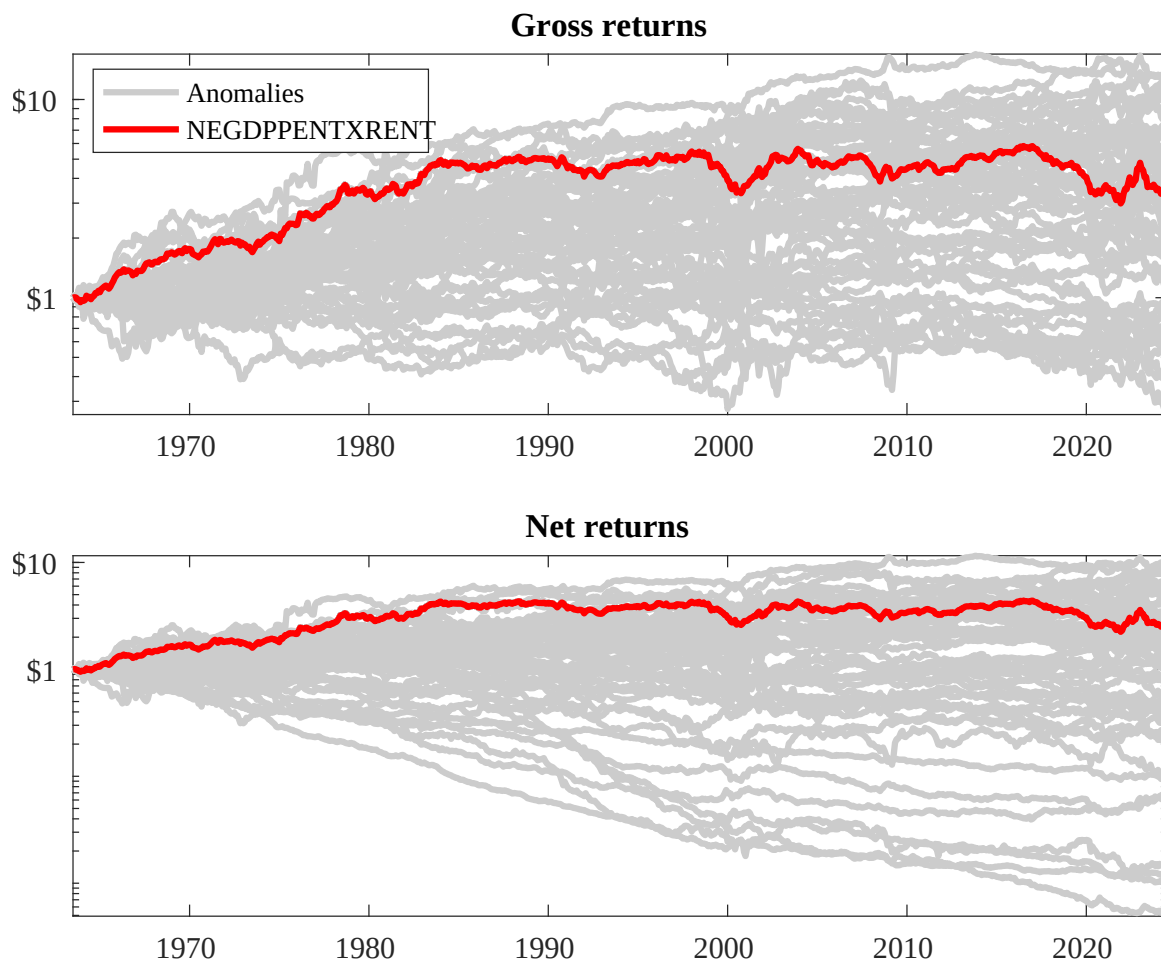
This table presents results for conditional double sorts on size and ALI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ALI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ALI and short stocks with low ALI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                                                    |                 |                |                  |                  |                  |                  |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|------------------|------------------|------------------|------------------|
| Size quintiles                                                        | ALI Quintiles |                |                |                |                | ALI Strategies                                     |                 |                |                  |                  |                  |                  |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$   | $\alpha_{FF5}$   | $\alpha_{FF6}$   |                  |
|                                                                       | (1)           | 0.47<br>[1.86] | 0.89<br>[3.46] | 0.93<br>[3.67] | 1.01<br>[3.82] | 0.97<br>[3.41]                                     | 0.50<br>[3.98]  | 0.49<br>[3.87] | 0.42<br>[3.37]   | 0.30<br>[2.40]   | 0.38<br>[3.05]   | 0.29<br>[2.31]   |
|                                                                       | (2)           | 0.70<br>[2.89] | 0.83<br>[3.48] | 0.80<br>[3.45] | 0.91<br>[4.05] | 0.96<br>[4.08]                                     | 0.27<br>[2.61]  | 0.29<br>[2.88] | 0.20<br>[2.01]   | 0.17<br>[1.72]   | 0.15<br>[1.45]   | 0.13<br>[1.29]   |
|                                                                       | (3)           | 0.63<br>[2.94] | 0.80<br>[3.67] | 0.77<br>[3.60] | 0.86<br>[4.11] | 1.02<br>[4.74]                                     | 0.40<br>[4.12]  | 0.40<br>[4.11] | 0.32<br>[3.37]   | 0.28<br>[2.93]   | 0.25<br>[2.66]   | 0.23<br>[2.39]   |
|                                                                       | (4)           | 0.57<br>[2.83] | 0.75<br>[3.73] | 0.77<br>[3.89] | 0.77<br>[3.97] | 0.82<br>[4.10]                                     | 0.26<br>[2.49]  | 0.23<br>[2.26] | 0.13<br>[1.29]   | 0.09<br>[0.89]   | -0.06<br>[-0.66] | -0.08<br>[-0.80] |
|                                                                       | (5)           | 0.54<br>[3.14] | 0.60<br>[3.61] | 0.54<br>[3.17] | 0.55<br>[3.18] | 0.59<br>[3.49]                                     | 0.05<br>[0.46]  | 0.05<br>[0.47] | -0.09<br>[-0.88] | -0.12<br>[-1.14] | -0.31<br>[-3.17] | -0.31<br>[-3.12] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                                                    |                 |                |                  |                  |                  |                  |
| Size quintiles                                                        | ALI Quintiles |                |                |                |                | ALI Quintiles                                      |                 |                |                  |                  |                  |                  |
|                                                                       | Average $n$   |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                  |                  |                  |                  |
|                                                                       | (L)           | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)              | (H)              |                  |                  |
|                                                                       | (1)           | 295            | 294            | 293            | 289            | 288                                                | 30              | 28             | 24               | 22               | 22               |                  |
|                                                                       | (2)           | 88             | 88             | 88             | 88             | 87                                                 | 47              | 48             | 48               | 47               | 46               |                  |
|                                                                       | (3)           | 65             | 65             | 65             | 65             | 65                                                 | 85              | 86             | 85               | 86               | 85               |                  |
|                                                                       | (4)           | 56             | 57             | 56             | 56             | 56                                                 | 186             | 192            | 194              | 188              | 192              |                  |
| (5)                                                                   | 53            | 53             | 53             | 53             | 53             | 1792                                               | 1760            | 1319           | 1349             | 1265             |                  |                  |



**Figure 2:** Distribution of Sharpe ratios.

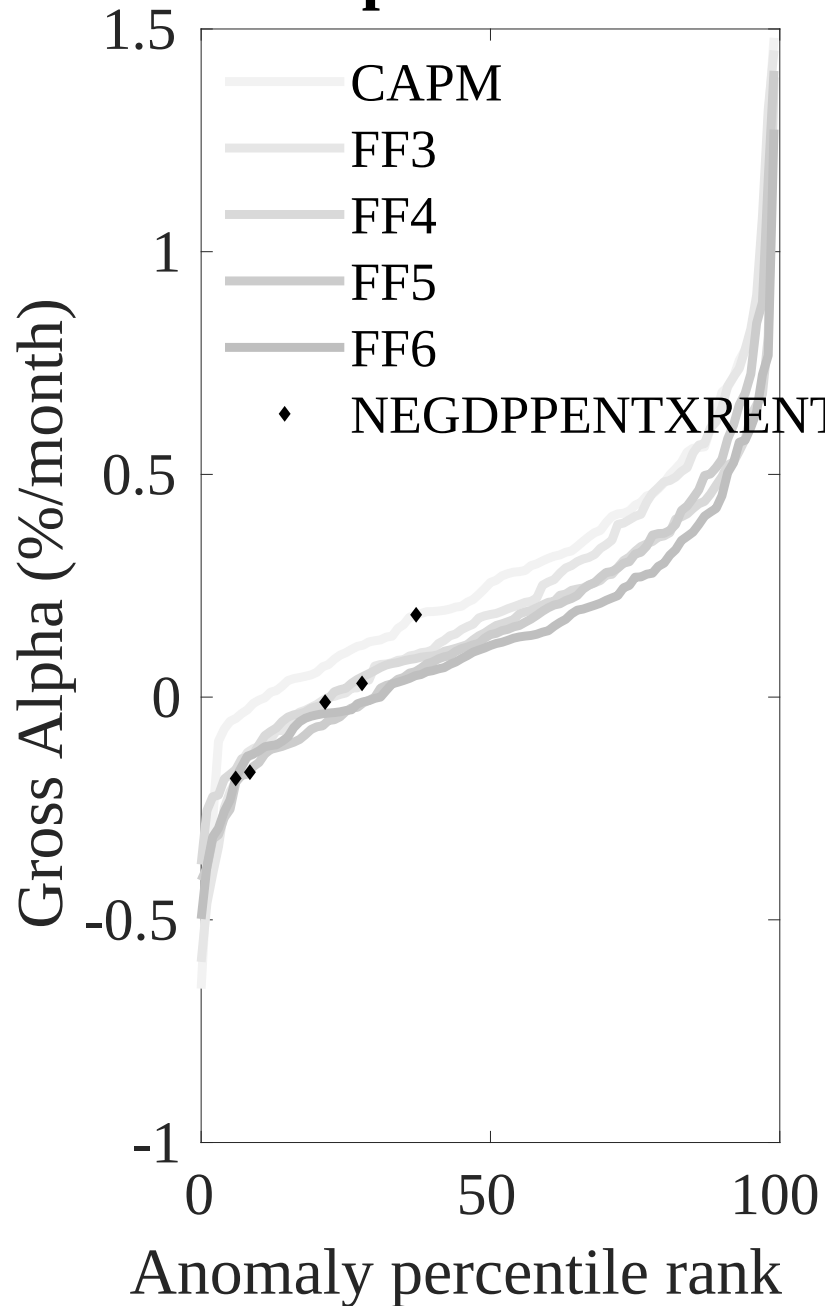
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ALI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

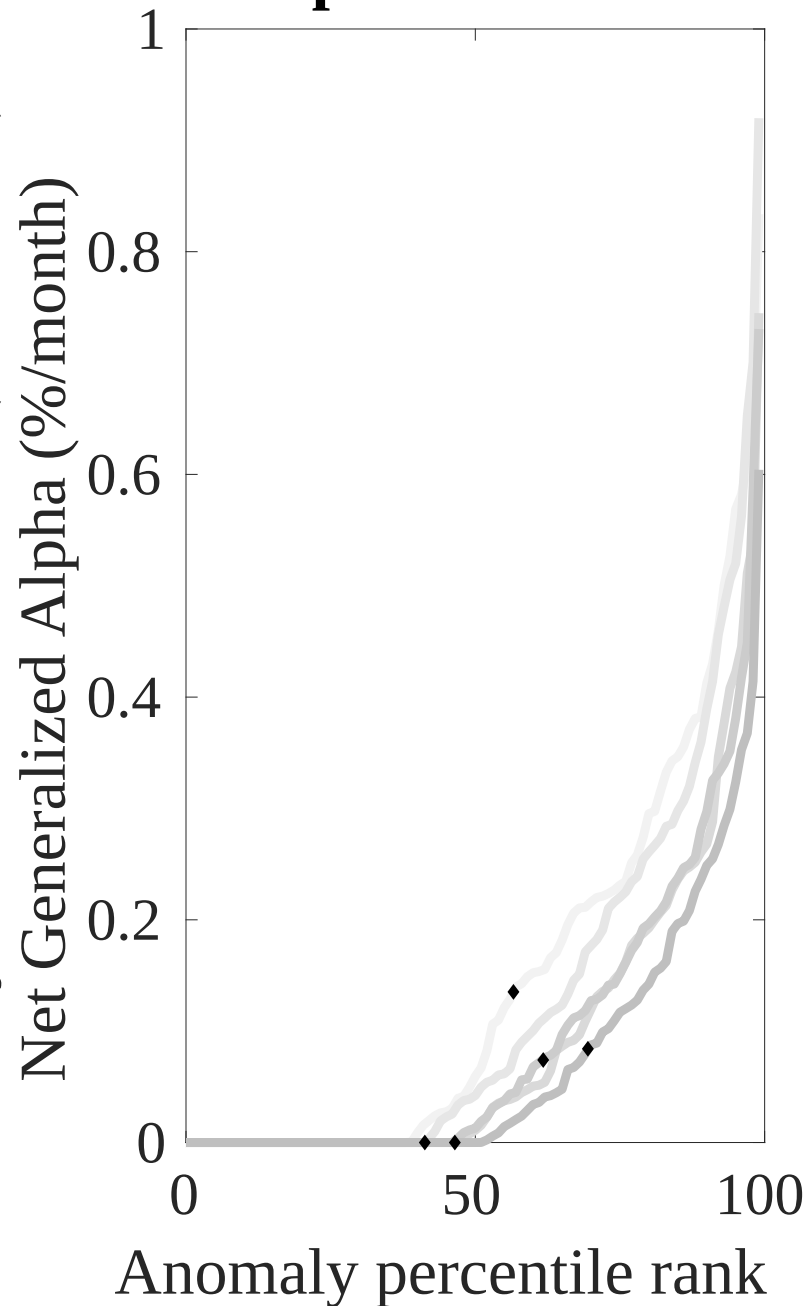
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ALI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

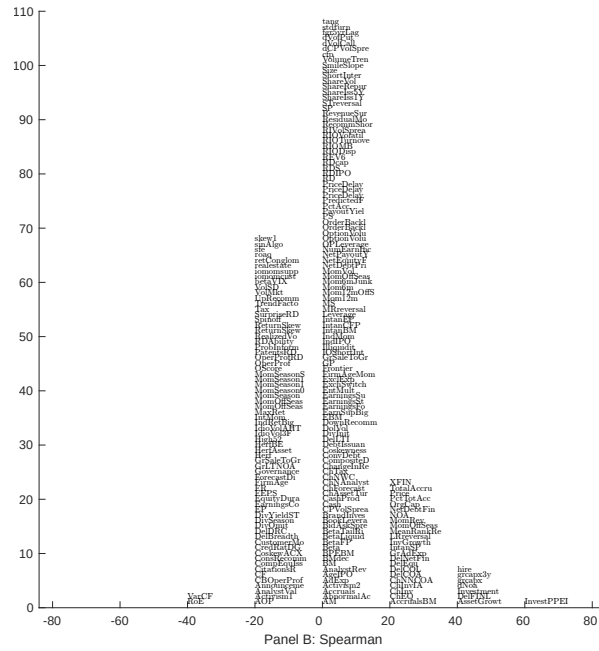
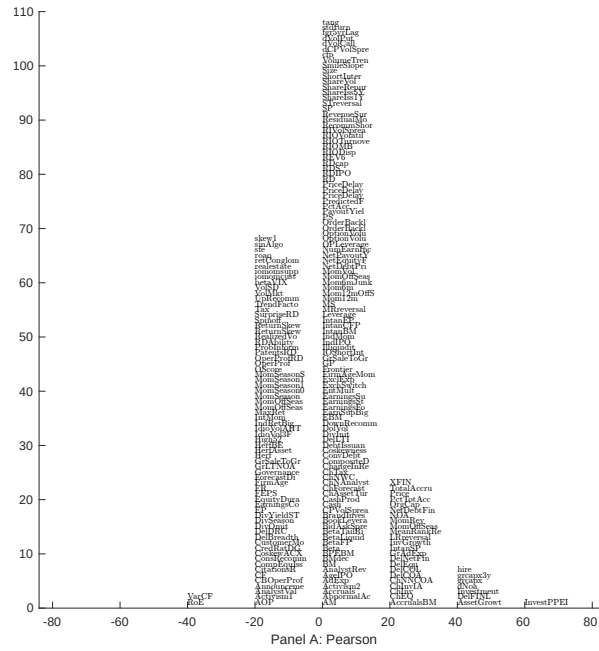
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

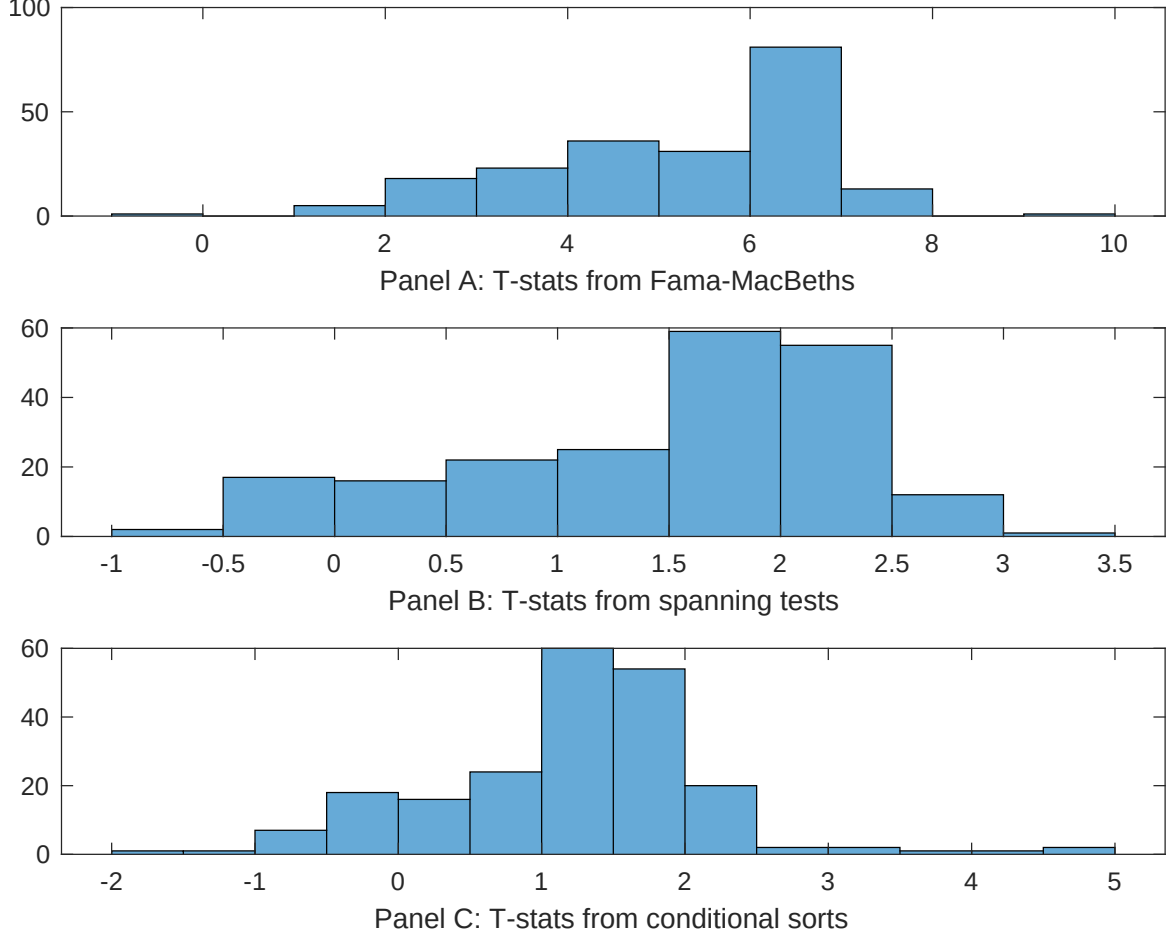
## Net Alpha Percentiles





**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with ALI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ALI conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{ALI}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{ALI,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ALI. Stocks are finally grouped into five ALI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ALI trading strategies conditioned on each of the 209 filtered anomalies.



**Table 4:** Fama-MacBeths controlling for most closely related anomalies

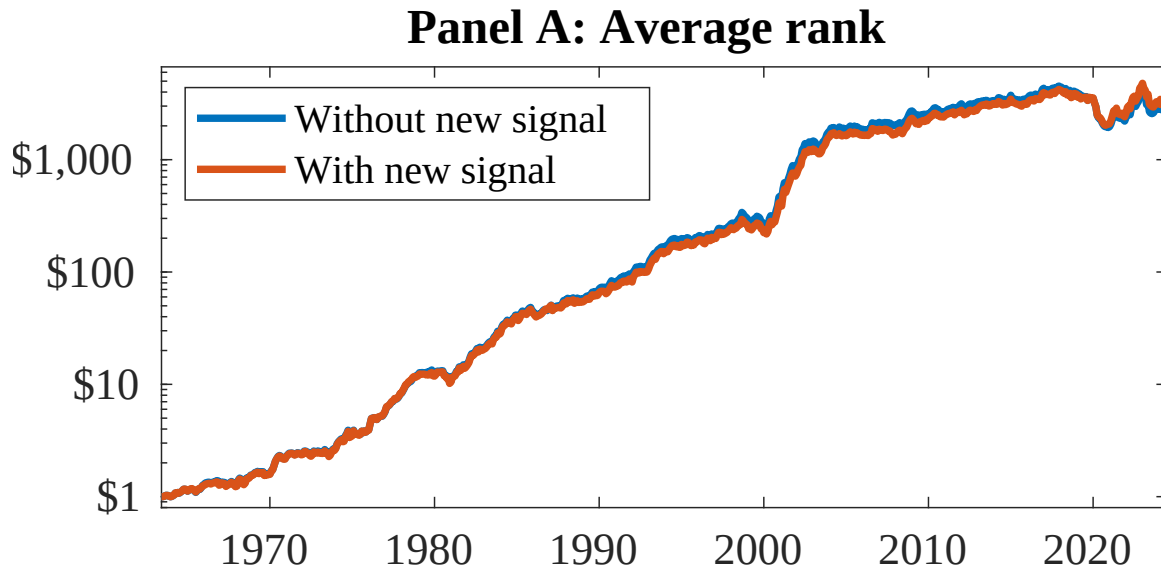
This table presents Fama-MacBeth results of returns on ALI. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{ALI}ALI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.14<br>[6.02] | 0.14<br>[6.56] | 0.13<br>[5.92] | 0.13<br>[5.87] | 0.14<br>[6.14] | 0.13<br>[5.51] | 0.14<br>[6.05]   |
| ALI             | 0.33<br>[2.54] | 0.49<br>[3.70] | 0.64<br>[4.79] | 0.44<br>[3.36] | 0.40<br>[3.04] | 0.71<br>[5.84] | 0.25<br>[1.88]   |
| Anomaly 1       | 0.14<br>[7.37] |                |                |                |                |                | -0.13<br>[-0.54] |
| Anomaly 2       |                | 0.12<br>[7.09] |                |                |                |                | 0.29<br>[1.15]   |
| Anomaly 3       |                |                | 0.55<br>[5.97] |                |                |                | 0.12<br>[0.85]   |
| Anomaly 4       |                |                |                | 0.11<br>[7.07] |                |                | 0.38<br>[1.89]   |
| Anomaly 5       |                |                |                |                | 0.93<br>[7.71] |                | 0.46<br>[2.50]   |
| Anomaly 6       |                |                |                |                |                | 0.80<br>[5.49] | 0.11<br>[0.81]   |
| # months        | 732            | 727            | 727            | 720            | 732            | 713            | 708              |
| $\bar{R}^2(\%)$ | 0              | 0              | 0              | 1              | 1              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ALI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{ALI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are change in ppe and inv/assets, Change in capex (three years), Change in capex (two years), change in net operating assets, Asset growth, Employment growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

|                 |                  |                  |                  |                  |                  |                  |                  |
|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| Intercept       | -0.18<br>[-2.36] | -0.15<br>[-1.89] | -0.17<br>[-2.09] | -0.24<br>[-2.99] | -0.19<br>[-2.24] | -0.12<br>[-1.45] | -0.18<br>[-2.44] |
| Anomaly 1       | 50.95<br>[14.42] |                  |                  |                  |                  |                  | 39.19<br>[9.92]  |
| Anomaly 2       |                  | 38.64<br>[9.77]  |                  |                  |                  |                  | 23.50<br>[3.58]  |
| Anomaly 3       |                  |                  | 33.98<br>[8.45]  |                  |                  |                  | -1.70<br>[-0.26] |
| Anomaly 4       |                  |                  |                  | 37.91<br>[8.23]  |                  |                  | 14.01<br>[2.89]  |
| Anomaly 5       |                  |                  |                  |                  | 17.24<br>[3.40]  |                  | 3.75<br>[0.75]   |
| Anomaly 6       |                  |                  |                  |                  |                  | 25.91<br>[5.62]  | 2.28<br>[0.52]   |
| mkt             | 10.27<br>[5.70]  | 10.32<br>[5.36]  | 10.61<br>[5.44]  | 11.08<br>[5.66]  | 10.92<br>[5.38]  | 11.21<br>[5.58]  | 10.04<br>[5.75]  |
| smb             | 27.93<br>[10.70] | 22.39<br>[7.77]  | 25.82<br>[9.02]  | 30.37<br>[10.74] | 28.59<br>[9.65]  | 31.15<br>[10.71] | 24.07<br>[8.93]  |
| hml             | 3.32<br>[0.95]   | 1.45<br>[0.38]   | 2.69<br>[0.70]   | 5.92<br>[1.56]   | 8.37<br>[2.13]   | 4.26<br>[1.07]   | -1.13<br>[-0.32] |
| rmw             | 24.82<br>[7.06]  | 22.62<br>[6.02]  | 22.08<br>[5.77]  | 25.42<br>[6.66]  | 24.87<br>[6.28]  | 25.59<br>[6.53]  | 23.92<br>[6.99]  |
| cma             | 22.04<br>[3.80]  | 40.42<br>[6.93]  | 42.47<br>[7.15]  | 32.86<br>[5.03]  | 40.34<br>[4.80]  | 38.33<br>[5.46]  | 1.71<br>[0.22]   |
| umd             | 1.92<br>[1.07]   | -0.26<br>[-0.14] | -0.48<br>[-0.25] | 1.22<br>[0.63]   | 2.56<br>[1.27]   | 0.57<br>[0.28]   | 0.35<br>[0.20]   |
| # months        | 732              | 728              | 728              | 732              | 732              | 713              | 709              |
| $\bar{R}^2(\%)$ | 49               | 42               | 41               | 40               | 36               | 38               | 53               |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ALI. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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